

## Project Design Phase-II Technology Stack (Architecture & Stack)

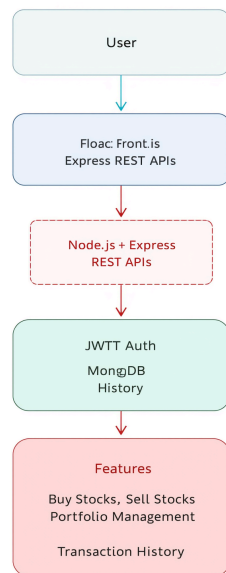
|               |                                 |
|---------------|---------------------------------|
| Date          | 31 January 3035                 |
| Team ID       |                                 |
| Project Name  | SHOPEZ : E-commerce Application |
| Maximum Marks | 4 Marks                         |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

### Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



**Table-1 : Components & Technologies:**

| S.No | Component           | Description                               | Technology                        |
|------|---------------------|-------------------------------------------|-----------------------------------|
| 1.   | User Interface      | Web application interface for users       | React.js, HTML5, CSS3, JavaScript |
| 2.   | Application Logic-1 | User Registration & Login Processing      | Node.js, Express.js               |
| 3.   | Application Logic-2 | Stock Trading Operations (Buy/Sell)       | Node.js, Express.js               |
| 4.   | Application Logic-3 | Portfolio & Transaction Management        | Node.js, Express.js               |
| 5.   | Database            | Stores users, portfolios and transactions | MongoDB                           |
| 6.   | Cloud Database      | Database hosting service                  | MongoDB Atlas                     |
| 7.   | File Storage        | Store application logs and assets         | Local Storage / Cloud Storage     |

|     |                                 |                                     |                                       |
|-----|---------------------------------|-------------------------------------|---------------------------------------|
| 8.  | External API-1                  | Stock Search and Market Information | Yahoo Finance API                     |
| 9.  | External API-2                  | Not Applicable for MVP              | Alpha Vantage API                     |
| 10. | Machine Learning Model          | Not Applicable for MVP              | N/A                                   |
| 11. | Infrastructure (Server / Cloud) | Application Hosting & Deployment    | Localhost / Render / Railway / Vercel |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                      | Technology                                           |
|------|--------------------------|--------------------------------------------------|------------------------------------------------------|
| 1.   | Open-Source Frameworks   | Frameworks used in application development       | React.js, Node.js, Express.js, MongoDB               |
| 2.   | Security Implementations | Authentication and data protection mechanisms    | JWT Authentication, Password Hashing (bcrypt), HTTPS |
| 3.   | Scalable Architecture    | Three-tier architecture supporting future growth | MERN Stack Architecture                              |

| S.No | Characteristics | Description                                        | Technology                    |
|------|-----------------|----------------------------------------------------|-------------------------------|
| 4.   | Availability    | Application accessible anytime through web browser | Vercel, Render, MongoDB Atlas |
| 5.   | Performance     | Fast API responses and optimized database queries  | Express.js, MongoDB Indexing  |

#### References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>